Singular Value Decomposition, Truncated PCA, and Reconstruction Error

Spectral Stability of Equity Covariance

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Abstract

For a centered data matrix, the singular value decomposition identifies the principal-component directions and expresses covariance eigenvalues in terms of squared singular values. We derive this correspondence directly, obtain the principal-component score matrix and its rank-k reconstruction, and calculate the exact Frobenius reconstruction error. The resulting formula also identifies the proportion of empirical variance retained by the first k principal components.

1 From the SVD to principal components

Let $X \in \mathbb{R}^{n\times d}$ be a centered data matrix: its rows are observations $x_1^\mathsf{T}, \dots, x_n^\mathsf{T}$ with $x_i \in \mathbb{R}^d$, and

$$\mathbf{1}_n^\mathsf{T} X = 0. \tag{1}$$

Assume $n \geq 2$ and $r = \operatorname{rank}(X) \geq 1$. The compact singular value decomposition is

$$X = UDV^\mathsf{T}, \qquad U \in \mathbb{R}^{n\times r}, \quad V \in \mathbb{R}^{d\times r}, \quad D = \operatorname{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}^{r\times r}, \tag{2}$$

where $\sigma_1 \geq \cdots \geq \sigma_r > 0$ and $U^\mathsf{T}U = V^\mathsf{T}V = I_r$. We write u_i and v_i for the columns of U and V, respectively.

With the normalization used here, the sample covariance matrix is

$$\widehat{\Sigma}_n = \frac{1}{n}X^\mathsf{T}X = \frac{1}{n}(UDV^\mathsf{T})^\mathsf{T}(UDV^\mathsf{T}) = \frac{1}{n}VD(U^\mathsf{T}U)DV^\mathsf{T} = V\frac{D^2}{n}V^\mathsf{T}. \tag{3}$$

Thus $\widehat{\Sigma}_n v_i = (\sigma_i^2/n)v_i$ for $i = 1, \dots, r$: the right singular vectors are principal-component directions, and the associated positive covariance eigenvalues are

$$\boxed{\lambda_i(\widehat{\Sigma}_n) = \frac{\sigma_i^2}{n}, \qquad i = 1, \dots, r.} \tag{4}$$

The remaining $d - r$ eigenvalues, when present, are zero.

Remark 1.1 (Why the columns of U are orthonormal). The relation $Xv_i = \sigma_i u_i$ gives $u_i = Xv_i/\sigma_i$. Because $X^\mathsf{T}Xv_j = \sigma_j^2 v_j$ and $v_i^\mathsf{T}v_j = \delta_{ij}$,

$$u_i^\mathsf{T}u_j = \frac{v_i^\mathsf{T}X^\mathsf{T}Xv_j}{\sigma_i\sigma_j} = \frac{\sigma_j^2}{\sigma_i\sigma_j}\,\delta_{ij} = \delta_{ij}.$$

Hence $U^\mathsf{T}U = I_r$, as required in the compact SVD.

Centering also constrains the rank. Equation (1) gives $X^\mathsf{T}\mathbf{1}_n = 0$, so X^T has a nonzero null vector. Therefore

$$\operatorname{rank}(X) \leq \min(d, n-1). \tag{5}$$

The denominator n in (3) corresponds to the empirical second-moment normalization. The usual sample covariance with denominator $n-1$ is unbiased under independent, identically distributed observations with finite second moments; its positive eigenvalues are $\sigma_i^2/(n-1)$.

2 Truncated SVD, scores, and reconstruction

Fix $1 \le k \le r$ and form the leading singular-vector blocks

$$U_k = [u_1, \dots, u_k] \in \mathbb{R}^{n\times k}, \qquad V_k = [v_1, \dots, v_k] \in \mathbb{R}^{d\times k}, \qquad D_k = \operatorname{diag}(\sigma_1, \dots, \sigma_k). \tag{6}$$

Proposition 2.1 (Principal-component scores)**.** *The matrix of the first k principal-component scores is*

$$\boxed{Z = XV_k = U_k D_k \in \mathbb{R}^{n\times k}.} \tag{7}$$

Its jth row consists of the coordinates $(x_j^\mathsf{T} v_1, \dots, x_j^\mathsf{T} v_k)$.

Proof. Starting with (2),

$$XV_k = UDV^\mathsf{T} V_k = UD \begin{pmatrix} I_k \\ 0 \end{pmatrix} = U \begin{pmatrix} D_k \\ 0 \end{pmatrix} = U_k D_k.$$

The row expression follows directly from matrix multiplication. □

Proposition 2.2 (Rank-k reconstruction)**.** *Let $P_k = V_k V_k^\mathsf{T}$ denote the orthogonal projector onto the first k PCA directions. The reconstruction of the centered data in this subspace is*

$$\boxed{\widehat{X}_k = XP_k = XV_k V_k^\mathsf{T} = U_k D_k V_k^\mathsf{T}.} \tag{8}$$

For each observation,

$$\widehat{x}_j = P_k x_j = \sum_{i=1}^{k} (x_j^\mathsf{T} v_i) v_i. \tag{9}$$

Proof. Substituting (7) into $\widehat{X}_k = XV_k V_k^\mathsf{T}$ gives

$$\widehat{X}_k = (XV_k)V_k^\mathsf{T} = U_k D_k V_k^\mathsf{T}.$$

Since V_k has orthonormal columns, $V_k V_k^\mathsf{T}$ is the orthogonal projector onto $\operatorname{span}(v_1, \dots, v_k)$, which gives (9). □

3 Reconstruction error and retained variance

The truncated SVD leaves out the singular directions with indices larger than k. We calculate its residual in the Frobenius norm, $\|A\|_F^2 = \operatorname{tr}(A^\mathsf{T} A)$.

Proposition 3.1 (Frobenius reconstruction error)**.** *For the reconstruction* (8)*,*

$$\boxed{\|X - \widehat{X}_k\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2.} \tag{10}$$

Proof. Using the Frobenius inner product $\langle A, B\rangle_F = \operatorname{tr}(A^\mathsf{T} B)$,

$$\|X - \widehat{X}_k\|_F^2 = \|X\|_F^2 + \|\widehat{X}_k\|_F^2 - 2\langle X, \widehat{X}_k\rangle_F. \tag{11}$$

The SVD and orthonormality yield

$$\|X\|_F^2 = \operatorname{tr}(X^\mathsf{T} X) = \operatorname{tr}(VD^2 V^\mathsf{T}) = \sum_{i=1}^{r} \sigma_i^2$$

and, similarly,

$$\|\widehat{X}_k\|_F^2 = \operatorname{tr}(V_k D_k^2 V_k^\mathsf{T}) = \sum_{i=1}^{k} \sigma_i^2.$$

For the cross term, using $\widehat{X}_k = XV_kV_k^\mathsf{T}$ and cyclicity of the trace,

$$\langle X, \widehat{X}_k\rangle_F = \operatorname{tr}(X^\mathsf{T}XV_kV_k^\mathsf{T}) = \operatorname{tr}(V_k^\mathsf{T}X^\mathsf{T}XV_k) = \operatorname{tr}(D_k^2) = \sum_{i=1}^{k}\sigma_i^2.$$

Substituting these three expressions into (11) gives

$$\|X - \widehat{X}_k\|_F^2 = \sum_{i=1}^{r}\sigma_i^2 + \sum_{i=1}^{k}\sigma_i^2 - 2\sum_{i=1}^{k}\sigma_i^2 = \sum_{i=k+1}^{r}\sigma_i^2.$$

□

Remark 3.2 (Eckart–Young–Mirsky theorem). The calculation proves the exact reconstruction error for the truncated SVD. The separate, classical Eckart–Young–Mirsky theorem establishes its *optimality*: among all matrices $Y \in \mathbb{R}^{n\times d}$ with $\operatorname{rank}(Y) \le k$, the truncated SVD attains the minimum Frobenius error, and that minimum is $\sum_{i=k+1}^{r}\sigma_i^2$. We use this established result without reproducing its general optimality proof.

The fraction of total empirical variance retained by the leading k components is

$$\boxed{\mathrm{EV}_k = \frac{\sum_{i=1}^{k}\lambda_i(\widehat{\Sigma}_n)}{\sum_{i=1}^{r}\lambda_i(\widehat{\Sigma}_n)} = \frac{\sum_{i=1}^{k}\sigma_i^2}{\sum_{i=1}^{r}\sigma_i^2}.} \tag{12}$$

Equivalently, (10) gives

$$\mathrm{EV}_k = 1 - \frac{\|X - \widehat{X}_k\|_F^2}{\|X\|_F^2}. \tag{13}$$

Thus retained variance and residual reconstruction error are complementary descriptions of the same truncated-SVD approximation.